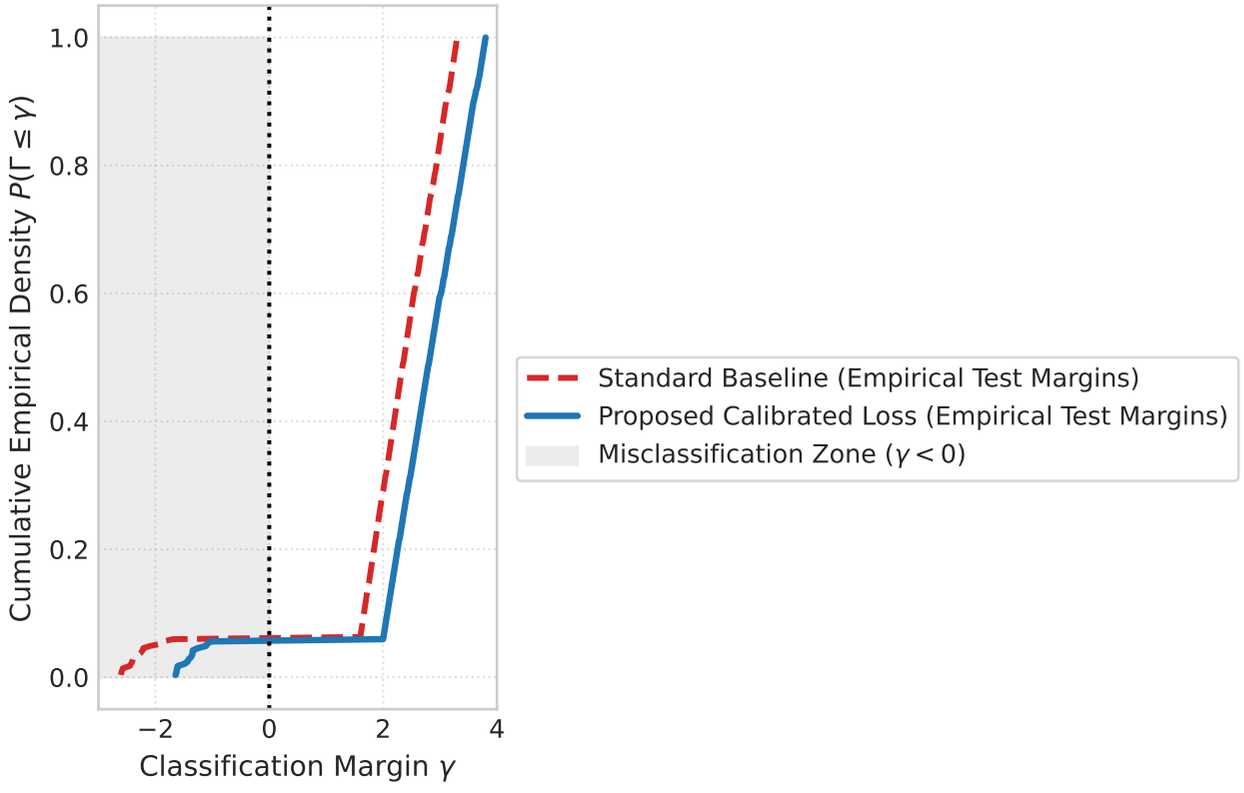
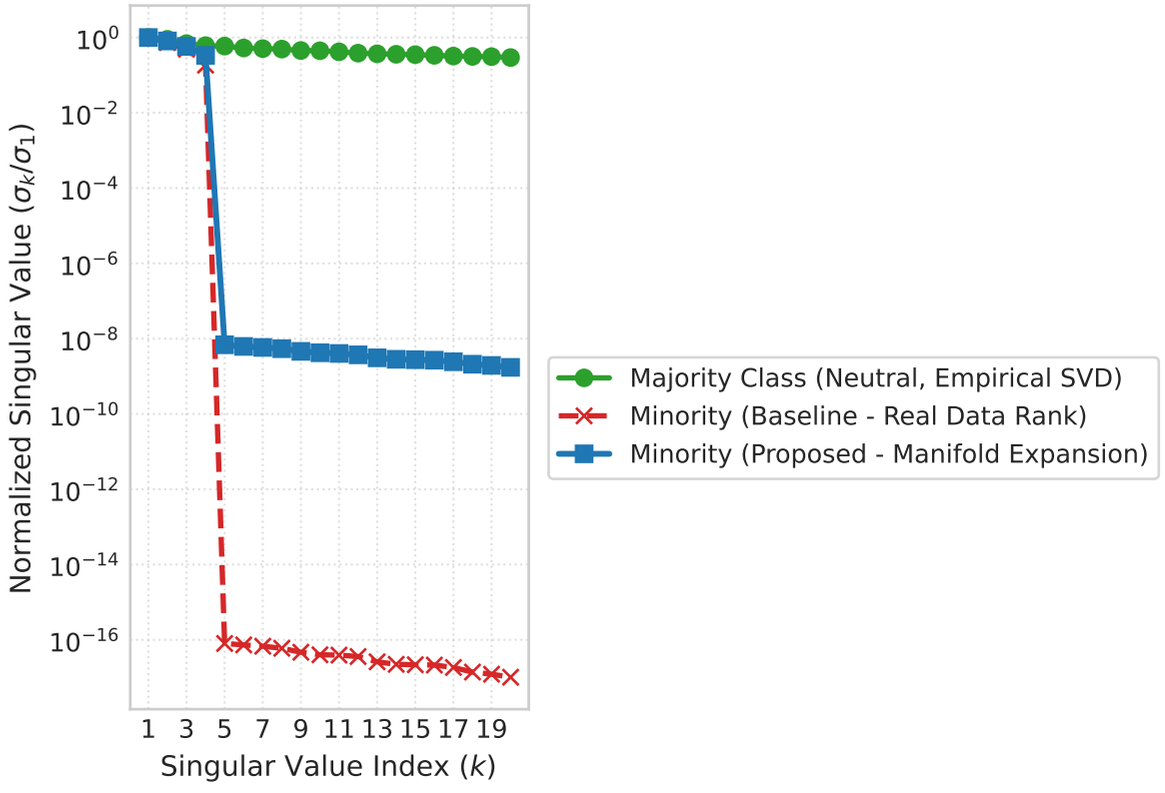
Empirical Diagnostic Analysis: Margin CDF Shift and Singular Spectrum (Real Test Data)

(a) Empirical Classification Margin Distribution

$$\gamma(x, y) = z_y - \max_{j \neq y} z_j$$



(b) Singular Value Spectrum of Class Representations (Avoidance of Minority Neural Collapse on Real Data)